MICROCIRCUITS, DIGITAL, BIPOLAR,
LOW-POWER SCHOTTKY TTL, OR GATES, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/305B, dated 9 August 1983, and is approved for use by all Department and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.2.3, outline letter: Delete "X" and substitute "2".

* 1.2.3, case outline; Add the following new case outline:

"X C-2A (20 terminal, .350" x .350"), square chip carrier package".

* 1.3, delete and substitute:

"1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	55 mW
Lead temperature (soldering, 10 seconds)	+300
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, C, D, X and 2	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	+175°C

1/ Must withstand the added P_D due to short circuit conditions (e.g., I_{OS}).

2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883."

PAGE 2

3.2.3, delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

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* The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
7	7
8	8

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PAGE 5

3.6, delete and substitute: "3.6 Marking. Marking shall be in accordance with MIL-M-38510."

TABLE II, test requirements column: Delete "Interim electrical parameters" and substitute "Initial electrical parameters".

4.2, subparagraphs b and c, delete and substitute:

"b. Initial and final electrical test parameters shall be as specified in table II, except initial electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

* TABLE II, additional electrical subgroups for group C periodic inspections; delete in its entirety.

* TABLE II, group A test requirements (method 5005), class B devices: Add "10, 11".

* 4.2a (1), delete "D or E" and substitute "D, E, or F".

PAGE 6

* 4.4.3c (3), delete and substitute the following:

"(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

* 4.4.3b: Delete in its entirety.

* 4.4.3c: Delete "4.4.3c" and substitute "4.4.3b".

PAGE 11

* TABLE III, pin designations: Delete "Case 2" and substitute "Cases 2, X".

PAGE 12

* TABLE III, pin designations: Delete "Case 2" and substitute "Cases 2, X".

* Footnote 1/: delete and substitute:

"1/ Cases 2 and X pins not referenced are NC."

PAGE 13

* TABLE III, pin designations: Delete "Case 2" and substitute "Cases 2, X".

PAGE 14

* TABLE III, pin designations: Delete "Case 2" and substitute "Cases 2, X".

PAGE 15

6.1: Delete in its entirety.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

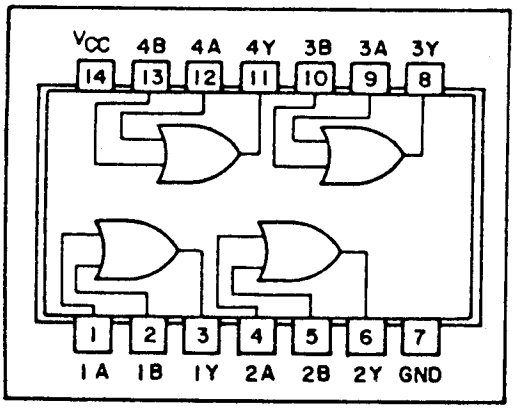
Agent:
DLA - ES

(Project 5962-1091)

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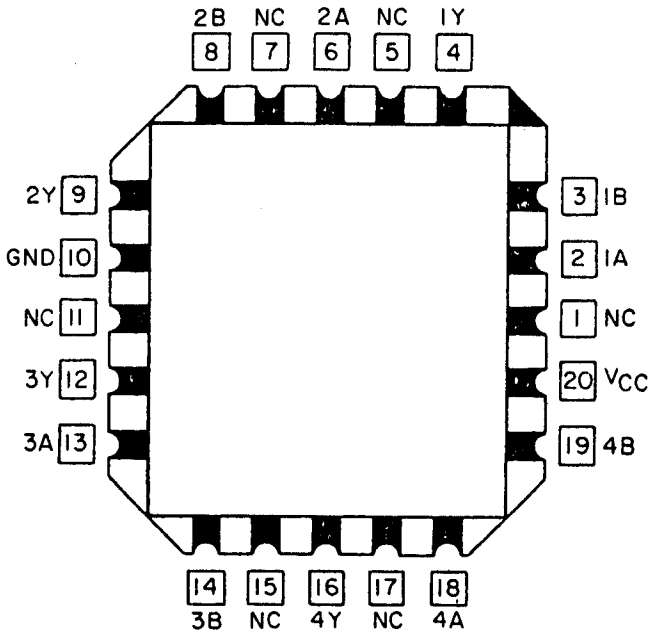
Device type 01

Cases A, B, C, and D



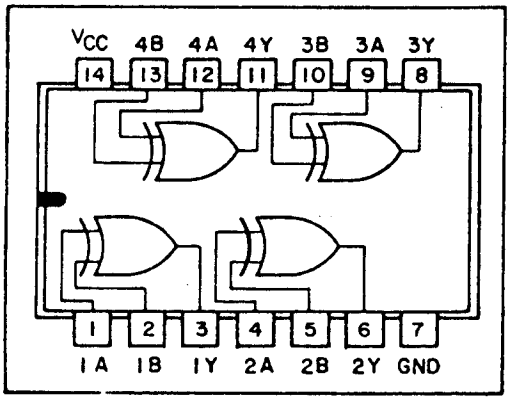
Device type 01

Cases 2 and X



Device type 02

Cases A, B, C, and D



Device type 02

Cases 2 and X

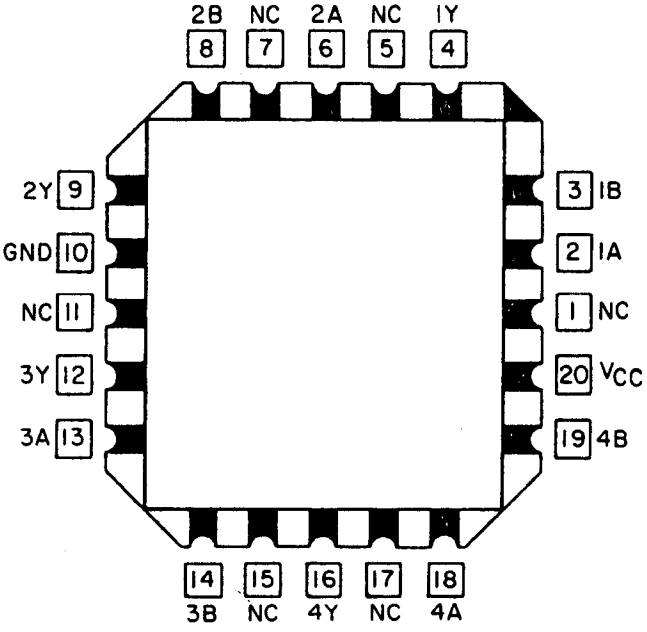


FIGURE 1. Logic diagram and terminal connections.

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Device type 01

Truth table (each gate)		
Inputs		Output
A	B	Y
H	X	H
X	H	H
L	L	L

X = Irrelvant
Positive logic: Y = A+B

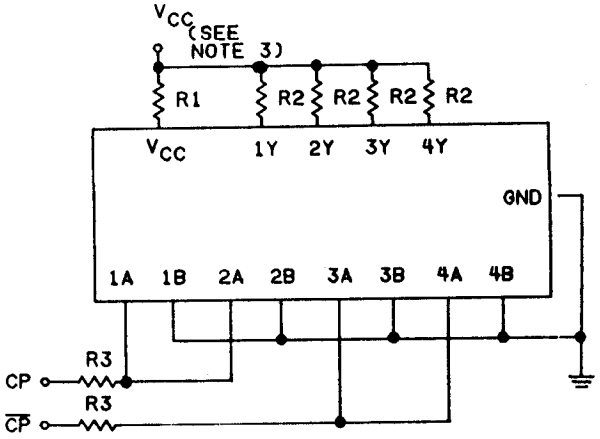
Device type 02

Truth table (each gate)		
Inputs		Output
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	L

Positive logic: Y = A+B

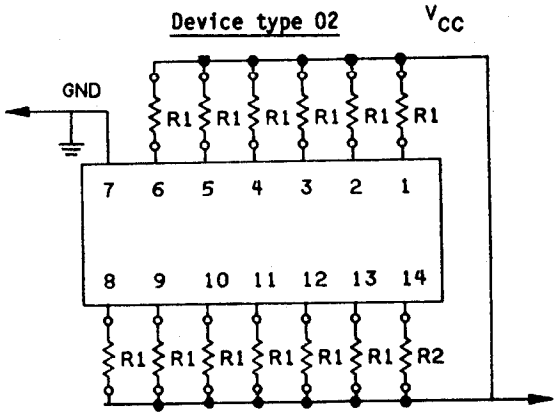
FIGURE 2. Truth tables.

Device types 01 and 02



NOTES:
1. CP or $\overline{CP}$ = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = 0.5 V minimum to 0.7 V maximum.
2. R_1 = 10Ω ±5 percent, R_2 = 1 kΩ ±5 percent, R_3 = 27Ω ±5 percent.
3. V_{CC} shall be high enough to insure that 5.0 V is present at device V_{CC} terminal.

Device type 02



High temperature burn-in

NOTES:
1. High temperature condition F (175°C) Condition F burn-in circuit only.
2. R_1 = 2 kΩ, R_2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuit for devices 01 and 02.

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